

VR-Forms. Preprint.

Part VII. Two-Register Logic

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VII.0. Introduction

The apparatus of VR-Forms was built in Parts II–IV; applications were given in Parts V–VI. The present part fixes the **logical properties** of the two-register system: which rules of inference act, how the two registers interact, whether classical logic is preserved, what peculiarities arise on the boundary.

Part VII is the technical closure of the system. No new ontological commitments are accepted here. The aim is to collect in one place the logical properties scattered through the preceding parts and to make their consequences visible.

Contents: classical logic in both registers (§VII.1); mixed formulas and rules of work with them (§VII.2); quantifiers in the formal register (§VII.3); the law of the excluded middle and its status (§VII.4); connection with two-sorted logic and other formal systems (§VII.5); summary (§VII.6).

VII.1. Classical logic in both registers

The principled decision, fixed in Part II §II.5 and justified by Theorem III.1: VR-Forms uses **classical first-order logic** in both registers.

What this means

In the ontological register the usual rules of classical logic act: modus ponens, generalisation, axiom schemes of classical predicate calculus. This is inherited from VR-Sets — there too classical logic (VR-Sets §I.3).

In the formal register — the same rules. The formal register is not a new logic. It is the same logical apparatus applied to a new type of objects (formal terms).

Why classical, not intuitionistic

A common solution in the foundations of mathematics for systems with operational ontology is to pass to intuitionistic logic (Brouwer, Bishop). VR-Sets has already declined this move: classical logic is preserved (see VR-Sets §VIII.3, comparison with constructivism).

The reason in VR-Sets: operational definiteness is already built into the closure principle. Every operational functionality gives a definite answer to every query; «a third» in the ontology is absent. So the law of excluded middle holds not as a logical axiom but as a property of the ontology.

In VR-Forms the situation is different: in the formal register there may be contradictory descriptions (Russell's class, the liar paradox). Here classical logic holds for another reason

— because formal terms are formulas, and working with them is working with formulas; standard logic applies to formulas as usual. Contradictions in formal terms do not generate global contradictions, since the formal register is conservative over the ontological (Part III).

Remark on another possibility

VR-Forms could be considered with intuitionistic or paraconsistent logic in the formal register. That would lead to a different system. The present work chooses classical logic for simplicity and for compatibility with classical mathematics during transits (Part IV). Alternative variants remain open directions.

VII.2. Mixed formulas

Formulas of VR-Forms may simultaneously contain ontological terms and formal ones. For instance: «every element of A satisfies ψ », where A is an operational set and ψ is a predicate referring to a formal term. Or: « $\ulcorner \tau \urcorner$ has an element in A », where $\ulcorner \tau \urcorner$ is formal and A is operational.

Rules for working with mixed formulas

Mixed formulas are admissible in L_1 by the principle of forms and the usual syntactic rules. Logical operations apply to them standardly.

In conducting a derivation, a mixed formula may be transformed either into a purely ontological one (if all formal terms are eliminable — for example, are operationally realisable), or remain mixed. There is no requirement to «squeeze out» formal terms — they may remain in the derivation as intermediate elements.

Translation π on mixed formulas

The mapping π from §III.2 applies to mixed formulas naturally: ontological terms remain unchanged, formal terms are translated into their defining predicates.

In particular, the formula $\ulcorner x \in \tau \urcorner \wedge x \in A$, where $\tau = \{y : \psi(y)\}$ and A is operational, translates into $\psi(x) \wedge x \in A$. We obtain a purely ontological formula. This is the mechanism of conservativity (§III.2).

Substantive significance

Mixed formulas are the principal working tool in the practice of VR-Forms. They allow one to formulate statements of the form «the operational object A satisfies the property described formally». This is the typical situation of transits: the formal register provides a convenient language for the property, the operational provides a concrete object.

Example: « $\wp_VR(\omega)$ is a subfamily of $\ulcorner \wp(\omega) \urcorner$ » — a mixed formula stating that the operational set of describable subsets of ω is a part of the formal term «classical $\wp(\omega)$ ». The statement is admissible in the formal register; in the ontological register it is equivalent to «every element of $\wp_VR(\omega)$ satisfies the predicate ‘is a subset of ω ’», which is trivially true.

VII.3. Quantifiers in the formal register

Quantifiers $\forall$ and $\exists$ applied to formal terms require a separate comment — they have specific content.

Quantifier $\exists$ over formal terms

The notation $\exists \tau \varphi(\tau)$ means: «there exists a formal term τ such that φ ». Since formal terms are syntactic records, this is equivalent to: «there exists a syntactically correct record of a description τ for which the formula $\varphi(\tau)$ is true».

The set of all formal terms is countable (by the finiteness of the alphabet and the inductive definition of terms). Hence $\exists$ -quantification over formal terms is, in essence, quantification over a countable set of syntactic objects. This does not generate uncountable collections, does not require choice over uncountable families, does not violate the countability of the universe of VR-Sets.

Quantifier $\forall$ over formal terms

The notation $\forall \tau \varphi(\tau)$ means: «for every formal term τ , φ holds». Analogously: quantification over a countable set of syntactic objects.

Remark: in this quantification participate **all** formal terms — including non-realisable, contradictory, mythological. A statement $\forall \tau \varphi(\tau)$ is a statement about all descriptions without exception. This is a strong statement, and in practice rarely encountered; more often the bounded quantification $\forall \tau (P(\tau) \rightarrow \varphi(\tau))$ is used for some predicate P distinguishing a class of formal terms.

Substantive sense

Quantifiers over formal terms make possible statements of the form «every description of a property has such-and-such structure», «there exists a formal term with such-and-such characteristic». These are meta-level statements — about the syntactic language itself.

Their applicability is limited by the fact that they speak about syntax, not about objects. The statement «there exists a formal term τ dragon

VII.4. The law of the excluded middle

Status of LEM in the ontological register

In the ontological register of VR-Sets the law of the excluded middle ($\varphi \vee \neg\varphi$) holds — an inheritance from the classical logic of VR-Sets. Substantively: every operational functionality gives a definite answer; there is no «middle».

Status of LEM in the formal register

In the formal register LEM also holds — since the logic is classical. However, its substantive significance requires caution.

The statement « $\ulcorner \text{dragon} \urcorner$ exists $\vee \neg(\ulcorner \text{dragon} \urcorner$ exists)» is true in the formal register by LEM. But «exists» here is formal (i.e. «the formal term $\ulcorner \text{dragon} \urcorner$ is»), not ontological. So the statement is trivial: the formal term $\ulcorner \text{dragon} \urcorner$, of course, is (we have recorded it); hence the disjunction is true, since its first part is true.

When someone says «a dragon exists or does not exist» in the ontological sense — this is in another register. In the ontological register the formula « $\exists x \text{ dragon}(x)$ » translates into a statement about the existence of an operational realisation of a dragon; its truth value is determined by the operational ontology (and in this case is false — no operational realisation).

Remark

The distinction of registers in the application of LEM removes many classical philosophical difficulties. «Does god exist?» in the formal register — yes, $\ulcorner \text{god} \urcorner$ is a formal term. In the ontological — no, no operational realisation in the VR-ontology. This is not a contradiction but a difference of semantic registers of the question.

VR-Forms does not claim that one of the answers is «more correct» than the other. It asserts that they answer **different questions**, and explicit distinction of registers helps not to confuse them.

VII.5. Connection with other formal systems

Let us collect the explicit comparisons of VR-Forms with systems known in logic.

Two-sorted logic

A standard extension of first-order logic in which variables are divided into sorts. VR-Forms is formally a two-sorted system (ontological sort, formal sort). Peculiarity: the formal sort includes the ontological (every operational set has a formal correlate), but not vice versa. This is an asymmetric embedding.

NBG (Bernays–Gödel)

An extension of ZFC with classes as second-sort objects. VR-Forms is structurally analogous: formal terms play the role of classes in NBG (large collections that are not operational sets). Difference: classes of NBG are objects (although of the second sort); formal terms of VR-Forms are formulas. This gives VR-Forms greater clarity regarding ontological status (the formal register makes no claim of ontology).

Internal Set Theory (Nelson)

An extension of ZFC separating standard and nonstandard objects. Structural similarity with VR-Forms: two registers, passage between them. Difference in motivation: IST introduces

the nonstandard as a tool for proofs about the standard; VR-Forms introduces the formal as a way to speak about the non-existent without ontology.

Furthermore: IST ontologically extends ZFC (nonstandard objects «are» in some sense); VR-Forms does not ontologically extend VR-Sets (formal terms are formulas, not objects). This is a principal distinction in the spirit of the position.

Constructive and intuitionistic logic

Constructivism (Brouwer, Bishop) and intuitionism (Heyting) reject LEM and non-reflexive proofs of existence. VR-Forms preserves classical logic. The difference is principled: constructivism refuses language for the non-constructive; VR-Forms permits the language but deprives it of ontological force.

One could say: VR-Forms gives classical logic in the formal register with operational ontology in the ontological. This removes the tension between «classical mathematics works» and «not all classical objects are real»: they work as formulas; real are those that are operationally realisable.

Realisability (Kleene, Kreisel)

Realisability connects classical and constructive mathematics: a formula is classically provable and constructively realisable if it has a constructive witness. The notion of operational realisability (Definition II.3) is an analogue of realisability in VR-Forms. The difference: realisability classically answers «can the proof be constructivised?»; operational realisability in VR-Forms answers «does the formal term have an operational correlate?»

Free logic (Lambert, Bencivenga)

Free logic admits terms not having reference (terms denoting the non-existent). VR-Forms is formally similar to free logic: formal terms are terms without reference. Difference: free logic is a single theory with a single register; VR-Forms has two registers, with strictly formalised passage between them.

VII.6. Summary of Part VII

Established:

- (1) Classical first-order logic acts in both registers of VR-Forms (§VII.1).
- (2) Rules for working with mixed formulas; the translation π applies to them naturally (§VII.2).
- (3) Quantifiers over formal terms are quantifiers over a countable set of syntactic objects; do not generate uncountability (§VII.3).
- (4) The law of the excluded middle holds in both registers; careful distinction of registers removes many classical difficulties with LEM for non-operational objects (§VII.4).
- (5) Connections of VR-Forms with two-sorted logic, NBG, IST, constructivism, realisability, free logic (§VII.5).

Methodological summary. VR-Forms is logically closed. Its inference rules are clear, the relations between registers are formalised, connections with known systems are established. The system makes no hidden logical assumptions beyond classical first-order logic and the principles of VR-Sets.

What follows. Part VIII — the concluding part — records open questions, indicates the programme of VR-Audit as a natural continuation, and gives the general summary of the VR cycle.